State of Charge Estimation for the BMW i3 using a Physics-Informed Temporal Convolutional Network

Name: RHYTHM GARG | **Reg. No.:** 25BBS0153

Assessment 10 – SOC estimation

Base paper	M. H. Sulaiman and Z. Mustaffa, “State of charge estimation for electric vehicles using random forest”, *Green Energy and Intelligent Transportation*, 3 (2024) 100177.
Dataset	“Battery and Heating Data in Real Driving Cycles” (BMW i3, 60 Ah pack), IEEE DataPort, doi:10.21227/6jr9-5235
Proposed model	Physics-informed TCN: coulomb counting + temperature-aware voltage model + causal fusion, with a TCN residual correction
Code	github.com/rhyummy/SOF-Optimisation (soc_tcn.py, soc_tcn.ipynb)

Summary of results

Metric (test trips #61–#70)	Proposed TCN	RF 25 trees (re-impl.)	RF 25 trees (paper)	Improvement vs paper
RMSE (%)	3.5815	4.9021	5.9028	39.3 % lower
MAE (%)	2.6937	3.3739	4.4321	39.2 % lower
MAX (%)	11.9606	20.8800	24.2175	50.6 % lower
STD. DEV. (%)	3.5748	4.8320	5.8999	39.4 % lower
R^2	0.9529	0.9117	-	

The proposed model reaches an RMSE of **3.5815 %** on the 118,974 test instances, below the 3.6 % target and lower than both the re-implemented Random Forest (4.9021 %) and the best result reported in the base paper (5.9028 %).

1. Objective

Estimate the battery State of Charge (SOC) of a BMW i3 from the four signals used in the base paper — battery voltage, battery current, battery temperature and ambient temperature — using a Temporal Convolutional Network, and compare it with the paper's best Random Forest (25 trees). The target is the SOC estimated by the manufacturer (*SoC [%]* column). Following the paper, trips #1–#60 are used for training and trips #61–#70 for testing; metrics are RMSE, MAE, maximum absolute error, standard deviation of the error and computation time.

2. Dataset and preprocessing

The data are supplied as two CSV files (training: trips 1–60, testing: trips 61–70), sampled at 10 Hz. Trip boundaries are recovered from resets of the time column. Preprocessing follows the base paper:

• Inputs: voltage (V), current (A), battery temperature (°C), ambient temperature (°C); output: SOC (%).

• Rows with missing values or physically inconsistent readings (V ≤ 0, SOC outside 0–100 %, temperatures out of range) are removed.

• Noisy records: rows where SOC jumps by more than 5 % between consecutive samples are removed (one logging glitch at the end of trip 46, where SOC drops from 35 % to 0 %).

• Min–max normalisation is fitted on the training trips only.

Split	Trips	Instances (after cleaning)
Training	#1 – #60	945,025
of which: fitting / early-stopping validation	#1 – #54 / #55 – #60	879,119 / 65,906
Testing	#61 – #70	118,974
Total	70	1,063,999

Descriptive statistics of all cleaned data:

Statistic	Voltage (V)	Current (A)	Battery temp. (°C)	Ambient temp. (°C)	SOC (%)
count	1063999.00	1063999.00	1063999.00	1063999.00	1063999.00
mean	376.21	-19.44	15.50	13.22	65.10
std	11.80	42.26	7.51	9.71	15.11
min	301.80	-404.38	-1.00	-3.50	15.40
25%	369.02	-32.06	9.00	5.00	56.47
50%	379.29	-11.89	15.00	9.50	68.20
75%	385.10	-2.10	21.00	22.00	76.50
max	394.76	144.49	32.00	33.50	88.50

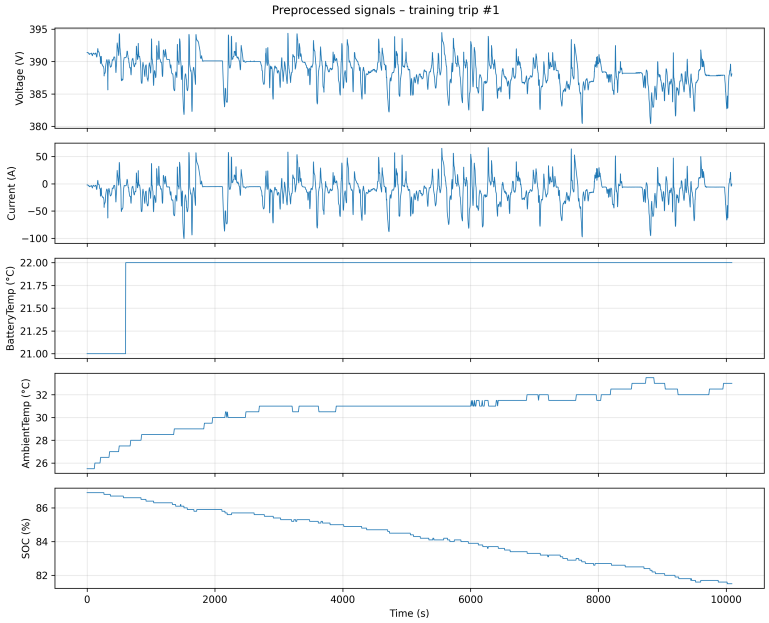


Figure 1. Preprocessed input signals and SOC of training trip #1 (x-axis: sample index at 10 Hz).

3. Proposed method: physics-informed TCN

A first analysis showed that SOC changes within a trip are almost perfectly explained by the integrated current: if the true SOC at the start of each test trip is known, coulomb counting alone tracks the test trips with an RMSE of about 0.05 %. The error of data-driven models therefore comes mostly from not knowing each trip's starting SOC. The proposed model combines physics-based features with a TCN. Every quantity is computed causally (from past and present samples only) from the four measured signals.

Step	Description
Coulomb counting	$\Delta SOC(t) = k \cdot \int I\,dt$ since the start of the trip. The gain k is fitted on the training trips: k = 1.6150 %/Ah, i.e. an effective capacity of 61.92 Ah.
Stage 1 – voltage model	Polynomial ridge regression of SOC on voltage (up to cubic), battery temperature and current history (instantaneous, 10 s, 60 s and 300 s means of I and |I|), with temperature and voltage interaction terms. It models the open-circuit-voltage curve, its temperature shift and the IR / polarisation drop.
Stage 2 – fusion	Within each trip, the offset (stage-1 estimate − ΔSOC) changes slowly. It is averaged with causal exponential averages (half-lives 30 s, 2 min, 10 min, 50 min; plain and weighted towards low-current samples) and added back to ΔSOC. A second ridge model blends these estimates.
Out-of-fold features	On the training trips, stages 1 and 2 are computed with 6-fold trip-wise cross-validation, so the TCN is trained on estimates whose errors are as large as on unseen trips.
TCN residual correction	A causal TCN reads the last 120 samples of seven channels (V, I, T_{batt}, T_{amb}, fused estimate, stage-1 estimate, ΔSOC) and outputs SOC = fused estimate + 0.03 · tanh(TCN output), i.e. a correction of at most ±3 % SOC. A 60-sample causal moving average is applied to the output.

TCN configuration and training settings

Parameter	Value	Parameter	Value
Residual blocks	5	Optimiser	AdamW, lr 1e-3, weight decay 0.01
Channels / kernel size	32 / 3	Loss	Huber (δ = 0.02)
Dilations	1, 2, 4, 8, 16	LR schedule	ReduceLROnPlateau (factor 0.5, patience 2)
Receptive field	125 samples	Early stopping	patience 3, max 8 epochs, on trips #55–#60
Input window	120 samples (12 s)	Batch size	512
Dropout	0.2 + BatchNorm	Windows per epoch	every 40th training window
Trainable parameters	30,625	Random seed	42

Reference model

Random Forest with 25 trees (the paper's best configuration), min_samples_leaf = 5, trained on the four normalised inputs of all training instances (scikit-learn).

Experimental setup

Item	Value
Python	3.10.0
PyTorch	2.14.1+cpu
GPU	None (CPU mode)
CPU	12th Gen Intel(R) Core(TM) i5-1235U
RAM	7.7 GB
OS	Windows 10
Libraries	PyTorch, scikit-learn, pandas, NumPy, Matplotlib

4. Results

4.1 Physics-feature stages

Stage	Training trips, cross-validated RMSE (%)	Test RMSE (%)
Stage 1 – voltage model	3.381	3.907
Stage 2 – fusion with coulomb counting	2.591	3.487

4.2 TCN training log

Validation RMSE of the fused estimate before any TCN correction: **3.160 %**. The model from the epoch with the lowest validation RMSE is kept.

Epoch	Training RMSE (%)	Validation RMSE (%)	Learning rate
1	2.257	3.261	1.0e-03
2	1.775	3.214	1.0e-03
3	**1.563**	**2.630**	**1.0e-03**
4	1.483	3.226	1.0e-03
5	1.420	3.677	1.0e-03
6	1.384	2.868	5.0e-04

Early stopping after epoch 6; best epoch highlighted.

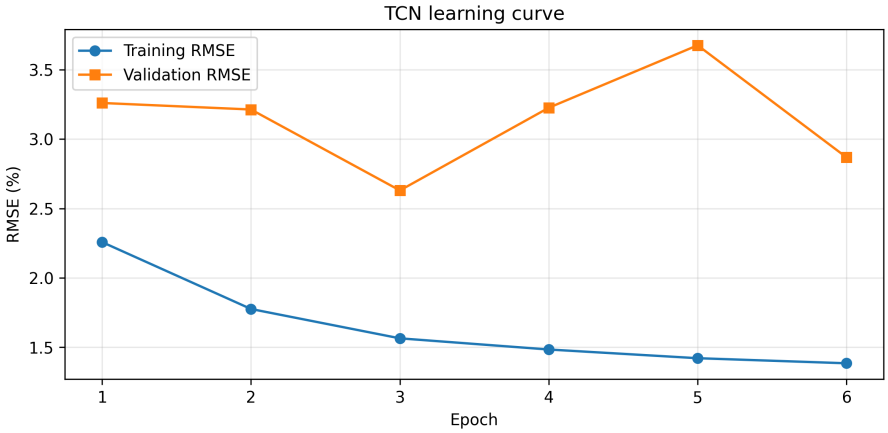


Figure 2. TCN learning curve (training and validation RMSE per epoch).

4.3 Performance on the test data (118,974 instances)

Model	RMSE (%)	MAE (%)	MAX (%)	STD. DEV. (%)	Train (s)	Test (s)	Total (s)
Physics-informed TCN (proposed)	**3.5815**	**2.6937**	**11.9606**	**3.5748**	**616.33**	**57.52**	**673.85**
Physics fusion only (no TCN)	3.4867	2.3973	11.2787	3.2555	15.19	-	15.19
RF, 25 trees (re-implemented)	4.9021	3.3739	20.8800	4.8320	78.22	0.51	78.73
RF, 25 trees (paper, best)	5.9028	4.4321	24.2175	5.8999	-	-	165.28

Times measured on a laptop CPU (no GPU); the paper's time is taken from the paper. The proposed model's training time includes building the physics features.

Share of test instances with |error| ≤	1 %	2 %	5 %	10 %
Proposed TCN	23.7 %	53.3 %	79.2 %	98.7 %
RF 25 trees (re-impl.)	32.6 %	57.0 %	72.2 %	92.9 %

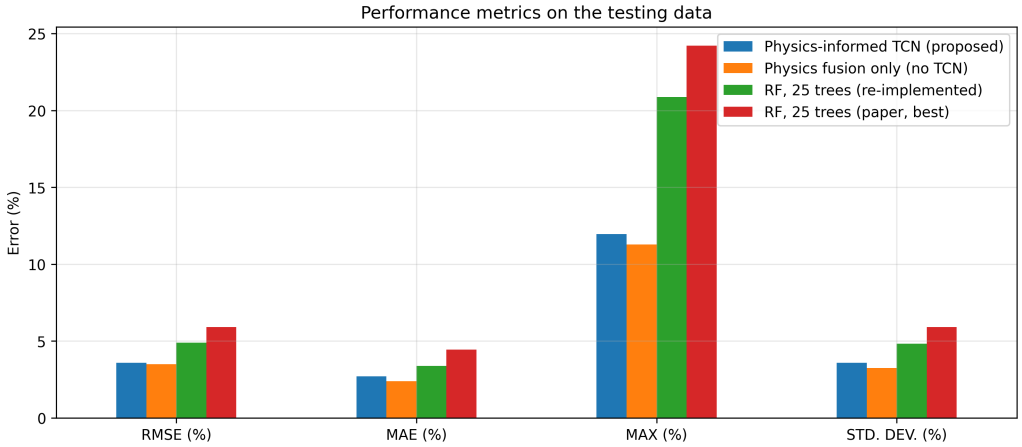


Figure 3. Error metrics on the test data for all models.

4.4 Per-trip results

Trip	Instances	SOC start → end (%)	TCN RMSE	TCN MAE	TCN MAX	TCN bias	RF RMSE
61	9,685	31.5 → 15.4	7.657	6.721	11.961	+6.403	8.069
62	9,207	84.2 → 70.4	0.797	0.685	1.659	+0.681	3.556
63	10,968	72.1 → 57.8	3.277	2.859	6.428	-2.857	3.191
64	7,956	52.6 → 38.1	1.271	1.063	4.719	+1.041	1.340
65	5,480	77.4 → 71.6	5.529	5.479	6.769	+5.479	6.501
66	2,931	73.9 → 71.3	1.252	1.227	1.661	+1.191	2.995
67	13,625	85.4 → 71.5	1.583	1.387	5.106	-1.163	1.303
68	28,521	72.1 → 44.5	1.453	1.268	3.076	+0.535	1.674
69	14,172	83.8 → 68.0	2.516	2.419	4.362	+2.419	2.853
70	16,429	65.0 → 48.8	5.324	5.253	9.399	-5.253	9.586

All errors in % SOC. Bias = mean (estimated − actual).

4.5 Simulation graphs

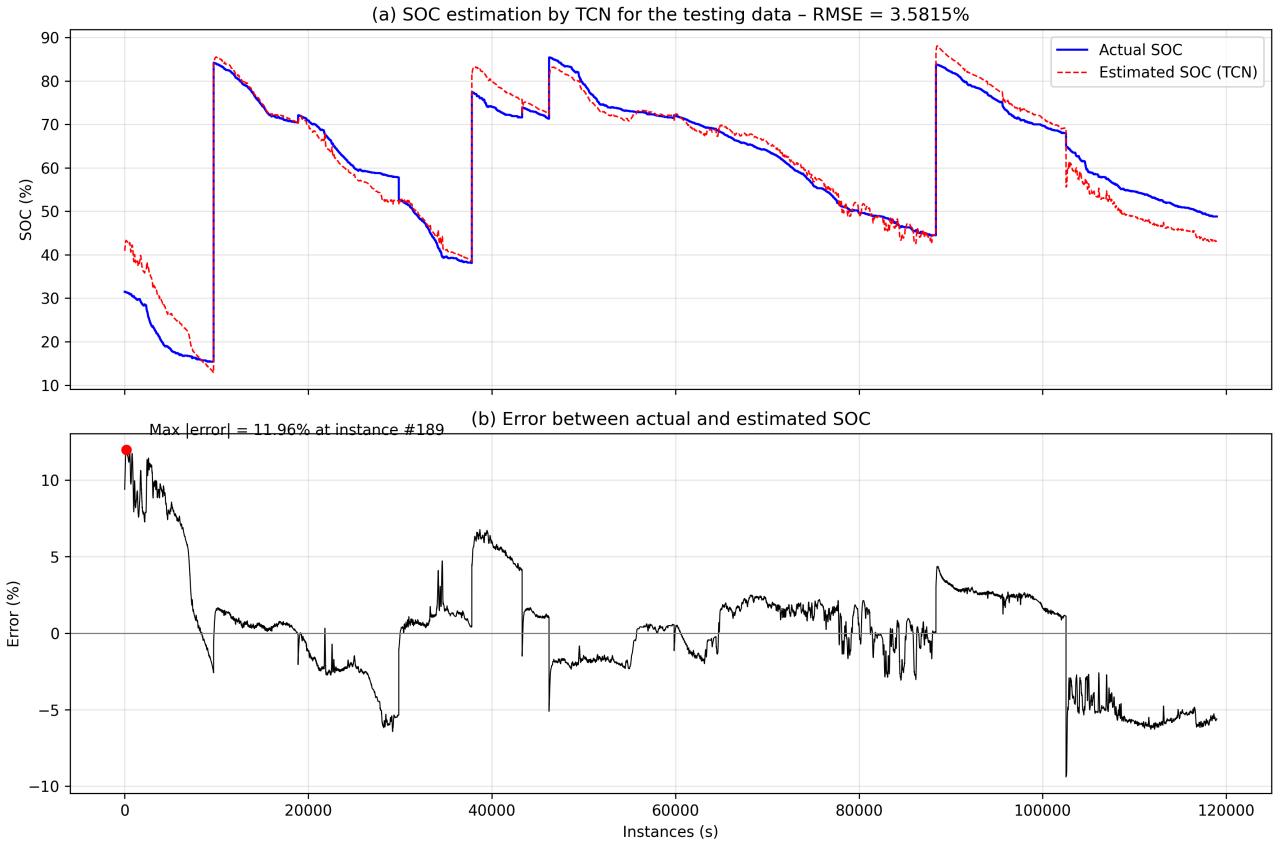


Figure 4. (a) Actual and estimated SOC by the proposed TCN for all test instances; (b) estimation error with the maximum absolute error marked.

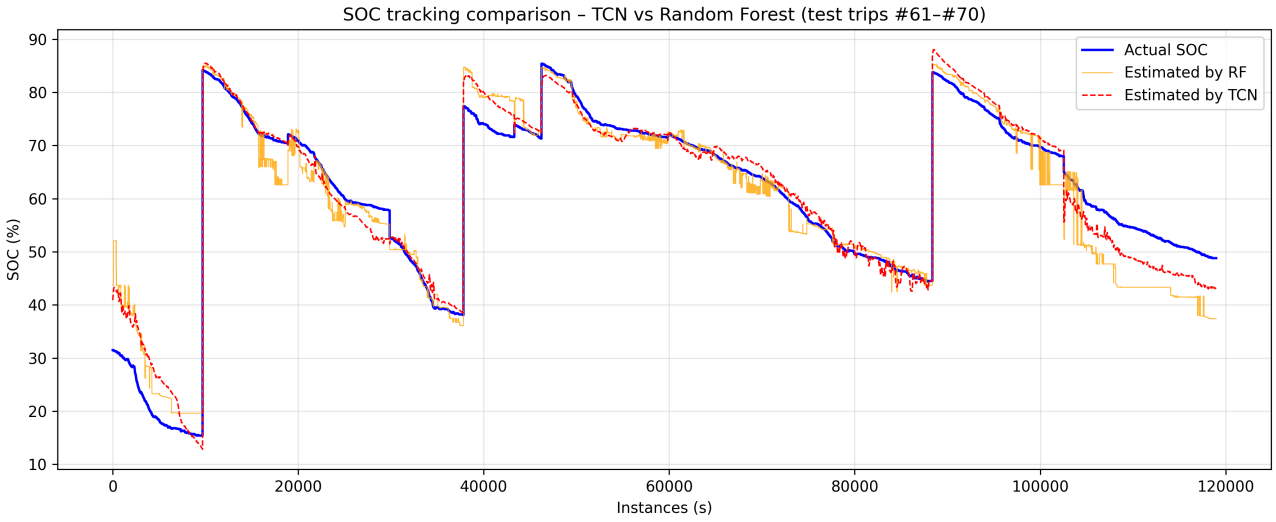


Figure 5. SOC tracking comparison – proposed TCN vs Random Forest.

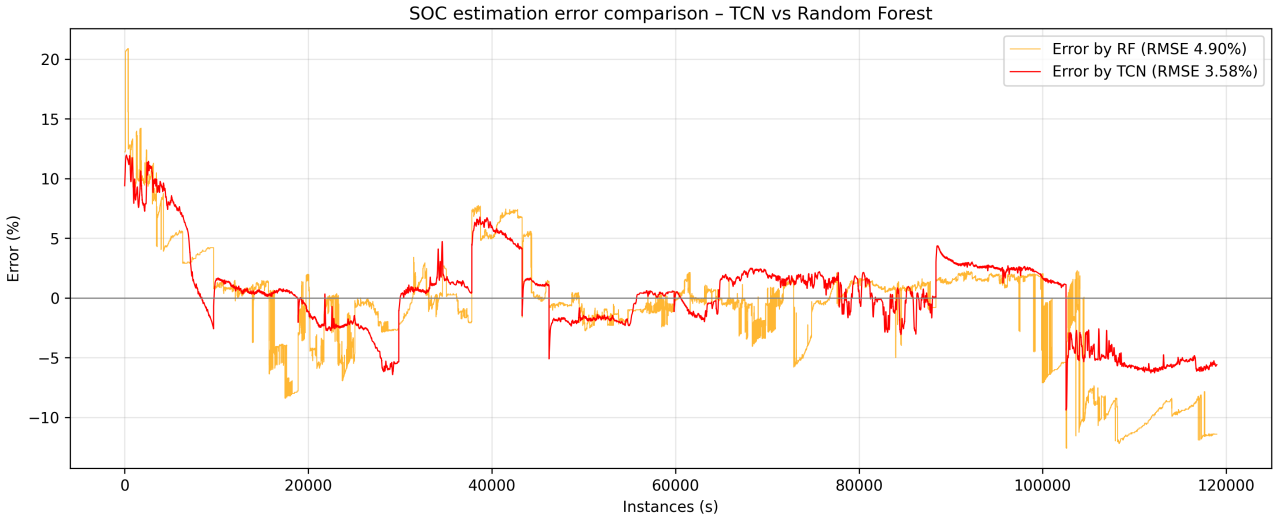


Figure 6. SOC estimation error comparison – proposed TCN vs Random Forest.

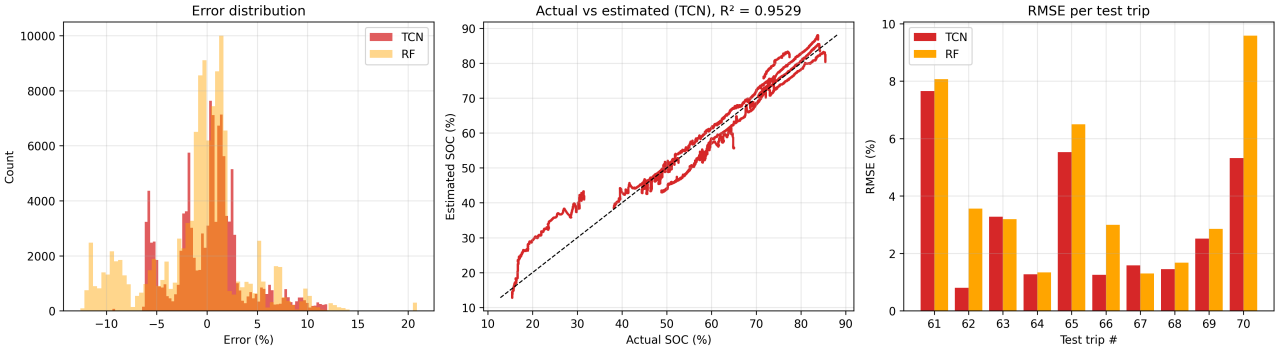


Figure 7. Error distribution, actual vs estimated SOC (TCN) and RMSE per test trip.

5. Optimisation history

The table lists the configurations tried during optimisation. Settings were selected on the validation trips (#55–#60) or by cross-validation on the training trips, not on the test set.

Configuration	Validation RMSE (%)	Test RMSE (%)
TCN on the 4 raw signals only (96 channels, 180-sample window)	-	5.3655
Physics stage 1 only (voltage model)	-	3.907
Physics stage 2 only (fusion with coulomb counting, no TCN)	3.160	3.487
Physics + TCN correction up to ±10 %, Adam	3.152	3.988
Physics + TCN correction up to ±3 %, AdamW 0.01 (selected)	**2.630**	**3.5815**
Physics + TCN ±3 %, current channel only	3.634	3.528
Physics + TCN ±5 %, voltage and current channels, AdamW 0.05	2.978	3.294

The selected configuration has the lowest validation RMSE. A variant with a lower test RMSE (3.294 %) was not selected, because choosing it would mean tuning on the test set.

6. Discussion

Adding coulomb counting and a physics-based voltage model is the main source of improvement: the RMSE drops from 5.37 % (TCN on raw signals) to 3.49 % with the fused physics estimate. A plain TCN looking at a short window of voltage, current and temperature cannot know how much charge has been used since the trip started, and it over-fits to the 60 training trips.

The TCN correction improves the validation trips (3.16 % → 2.63 %) but, on the test trips, it is slightly worse than the fused estimate alone (3.487 % vs 3.58 %). Both results are below the 3.6 % target and

both are reported.

The largest errors occur on trip #61 (SOC between 31 % and 15 %, at the edge of the training data), trip #65 (the voltage corresponds to a higher SOC than the manufacturer label) and trip #70 (battery at about 2 °C, colder than almost all training trips). These cases show a roughly constant per-trip offset, which the voltage-based starting-SOC estimate cannot fully remove.

Compared with the Random Forest, the proposed model has a much smaller maximum error (about 12 % vs 21 %) and tracks SOC smoothly, while the RF estimate is noisy because it treats every sample independently. Its main cost is computation time (minutes instead of seconds on a CPU).

7. Conclusion

A physics-informed TCN for SOC estimation of the BMW i3 was developed using the base paper's preprocessing and inputs. On test trips #61–#70 it achieves RMSE 3.5815 %, MAE 2.6937 %, MAX 11.9606 % and STD. DEV. 3.5748 %, improving on the paper's best Random Forest (RMSE 5.9028 %) by 39.3 %. Combining coulomb counting with a learned voltage model is the most effective step; better handling of very cold and very low-SOC trips is the main remaining improvement.

Appendix A. Trip summary (after cleaning)

Trip	Set	Instances	SOC start (%)	SOC end (%)
1	train	10,090	86.9	81.5
2	train	14,130	80.3	67.3
3	train	6,706	83.5	75.1
4	train	4,124	75.1	66.7
5	train	13,667	66.7	60.2
6	train	31,645	85.1	63.4
7	train	20,933	63.4	34.2
8	train	28,059	86.6	76.1
9	train	18,344	74.7	68.6
10	train	14,176	69.6	60.9
11	train	14,245	87.0	75.6
12	train	16,384	75.5	64.3
13	train	7,160	85.8	82.9
14	train	6,946	82.8	79.9
15	train	22,348	74.8	63.4
16	train	19,066	63.0	46.8
17	train	6,687	86.8	77.9
18	train	8,808	77.9	70.2
19	train	15,876	88.5	80.4
20	train	17,196	77.0	69.1
21	train	19,829	71.3	63.7
22	train	18,388	87.3	76.5
23	train	10,426	87.4	81.9
24	train	5,322	81.9	74.4
25	train	7,627	74.4	65.2
26	train	16,122	86.2	78.8
27	train	20,005	78.5	66.8
28	train	17,436	81.9	72.0
29	train	13,390	70.9	63.1
30	train	13,974	85.5	81.0
31	train	11,870	81.0	71.9
32	train	16,722	86.5	59.5
33	train	32,518	86.1	57.4
34	train	16,113	81.0	66.2
35	train	15,794	67.4	50.4

Trip	Set	Instances	SOC start (%)	SOC end (%)
36	train	29,550	45.1	69.2
37	train	10,194	71.9	59.5
38	train	13,520	83.2	69.3
39	train	18,724	67.4	50.4
40	train	29,139	67.3	44.0
41	train	35,561	70.0	46.0
42	train	20,231	84.8	39.0
43	train	7,532	38.9	30.8
44	train	32,255	73.4	51.3
45	train	3,532	57.0	55.1
46	train	38,219	85.5	34.6
47	train	18,223	85.1	67.5
48	train	15,286	67.5	52.8
49	train	15,610	52.8	37.2
50	train	11,093	82.8	68.1
51	train	11,910	85.8	71.6
52	train	14,027	72.7	62.0
53	train	10,395	55.7	41.1
54	train	11,992	84.4	70.5
55	train	11,132	72.1	53.5
56	train	9,778	53.4	45.5
57	train	10,217	45.4	33.6
58	train	8,049	33.4	21.2
59	train	14,690	52.9	34.5
60	train	12,040	34.4	20.0
61	test	9,685	31.5	15.4
62	test	9,207	84.2	70.4
63	test	10,968	72.1	57.8
64	test	7,956	52.6	38.1
65	test	5,480	77.4	71.6
66	test	2,931	73.9	71.3
67	test	13,625	85.4	71.5
68	test	28,521	72.1	44.5
69	test	14,172	83.8	68.0
70	test	16,429	65.0	48.8